

Sırrı Kaan Oğuzkan

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EDUCATION

- **Bilkent University** Ankara, Turkey
B.Sc. in Computer Engineering Sep 2023 – Present

EXPERIENCE

- **JotForm** Ankara, Turkey
Software Engineering Intern June 2026 - August 2026
 - **Element Registry Architecture:** Designed a plugin-contract architecture for a drag-and-drop email builder (**React 18**, **TypeScript**, **Redux**) where **20** content-block types share one typed interface across the canvas, panel, preview, and exporter, so new blocks required touching **zero** existing consumers.
 - **Bulk Send Pipeline:** Built an asynchronous bulk-email pipeline (PHP, MySQL, Redis) with a cron worker delivering via chunked SMTP sessions (**25** recipients/batch) and a **900s** timeout-based reclaim mechanism for stalled batches.
- **Alkan Lab for Bioinformatics and Computational Genomics** Ankara, Turkey
Undergraduate Volunteer February 2026 - Present
 - **CuCLARK GPU & CUDA Modernization:** Forked and modernized CuCLARK (GPU metagenomic classifier, originally CUDA 7.5 / single-arch); upgraded to **CUDA 11.8** with multi-arch support spanning **10** GPU generations (Kepler **sm_35** through Hopper **sm_90**, including K80, V100, A100, RTX 30xx/40xx, H100).
 - **Build System & Containerization:** Replaced legacy `install.sh` with a **CMake 3.20** build (C++17, CUDA 14, optional OpenMP) and authored a multi-arch `Dockerfile` published as `alkanlab/cuclark`; added a unified CLI with **6** subcommands wrapping the original scripts.
 - **Batch Classification Pipeline:** Diagnosed a byte-offset batching bug causing duplicate read assignments on small inputs and built a list-mode pipeline with resume, crash-recovery blacklist, and progress monitoring to classify **3000+** FASTA files in a single DB load.
- **Look & Cash** Ankara, Turkey
Part Time Software Engineer February 2025 - October 2025
 - **Ticket-Based Support System:** Built a ticket workflow for web/mobile (priority, comments, attachments, statuses). Shipped **13** REST endpoints, **4** ticket states, and **5** admin actions; reduced average admin handling time by **25%** via bulk actions and better filtering.
 - **Real-Time Mobile Support Chat:** Implemented real-time ticket chat on mobile with backend persistence and state sync; achieved **300 ms** p95 delivery and **99.5%** success under **150** concurrent sessions.
 - **Reporting & Graph Generation:** Built KPI reporting APIs over MongoDB (tickets, campaigns, scans, notifications) with date-range and campaign filters. Generated **14** dashboard datasets powering **4** chart types; cut report generation time from **6.2 s** to **1.9 s** via indexing.

PROJECTS

- **osguide.org — Interactive OS Learning Platform:** Built an interactive platform (**React**, **TypeScript**, **Vite**) structured around Bilkent's CS342 curriculum.
 - **Concept-to-Simulator Layers:** Designed a three-layer learning flow per topic — concept explanation, animation, and interactive simulator — covering core OS topics such as scheduling, memory management, and synchronization.
- **NutriApp — Meal Recommendation Platform:** Built backend services for a meal-planning app under the Amazon University Engagement Program.
 - **Serverless REST APIs:** Implemented **16** REST endpoints on AWS Lambda; won the **People's Choice Award**, ranking **1st of 7** teams.

SKILLS

- **Languages:** C++, TypeScript, JavaScript, PHP, Java
- **Frameworks & Tools:** React, Redux, Node.js, AWS Lambda, CUDA, Docker, Git
- **Databases:** MongoDB, DynamoDB, MySQL, Redis